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Inventor(s)	Sharifi; Matthew et al.

Shared assistant profiles verified via speaker identification

Abstract

A method for sharing assistant profiles includes receiving, at a profile service, from an assistant service interacting with a user device of a user, a request requesting the profile service to release personal information associated with the user to the assistant service. The operations also include performing, through the assistant service, a verification process to verify that the user consents to releasing the requested personal information by: instructing the assistant service to prompt the user to recite a unique token prescribed to the user; receiving audio data characterizing a spoken utterance captured by the user device of the user; processing the audio data to determine whether a transcription of the spoken utterance recites the unique token; and when the transcription of the spoken utterance recites the unique token, releasing, to the assistant service, the requested personal information stored on a centralized data store managed by the profile service.

Inventors:	Sharifi; Matthew (Kilchberg, CH), Carbune; Victor (Zürich, CH)
Applicant:	Google LLC (Mountain View, CA)
Family ID:	80222618
Assignee:	Google LLC (Mountain View, CA)
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Primary Examiner: Lagor; Alexander

Assistant Examiner: Binczak; Brandon

Attorney, Agent or Firm: Honigman LLP

Background/Summary

CROSS-INFERENCE TO RELATED APPLICATIONS (1) This U.S. patent application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application 63/264,167, filed on Nov. 16, 2021. The disclosure of this prior application is considered part of the disclosure of this application and is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to shared assistant profiles verified via speaker identification.

BACKGROUND

(2) Users frequently interact, with voice-enabled assistant interfaces on smart devices such as: without limitation, phones, watches, and smart speakers/displays. These assistant interfaces enable users to get things done and find answers to questions they might have, all through natural, conversational interactions. Developers are increasingly developing voice-enabled assistant interfaces for applications and services to permit user interaction through natural conversation. For example, automatic speech recognition (ASR) and natural language understanding (NLU) models may recognize and interpret queries spoken by users and fetch responses to these spoken queries. As users begin to interact with many different assistant services, it can become an annoyance to require users to input common details pertaining to user information repeatedly across the different assistant services yet still keep the user information private/secure.

SUMMARY

(3) One aspect of the disclosure provides a computer-implemented that when executed on data processing hardware causes the data processing hardware to perform operations that include receiving, at a profile service executing on the data processing hardware, from an assistant service interacting with a user device of a user, a request requesting the profile service to release personal information associated with the user to the assistant service. The operations also include performing, through the assistant service, a verification process to verify that the user consents to releasing the requested personal information to the assistant service by instructing the assistant service to prompt the user to recite a unique token prescribed to the user, receiving audio data characterizing a spoken utterance captured by the user device of the user; processing the audio data to determine whether a transcription of the spoken utterance recites the unique token; and when the transcription of the spoken utterance recites the unique token, releasing, to the assistant service, the requested personal information stored on a centralized data store managed by the profile service.

(4) Implementations of the disclosure may include one or more of the following optional features. In some implementations, the operations also include, while performing the verification process through the assistant service, processing the audio data to determine whether the spoken utterance captured by the user device was spoken by the user associated with the requested personal information. Here, releasing the requested personal information to the assistant service further includes releasing the requested personal information to the assistant service when the transcription

of the spoken utterance recites the unique token and the utterance was spoken by the user.

(5) The personal information requested by the assistant service may pertain to personal information required by the assistant service in order to fulfill a query submitted by the user to the assistant service via the user device. In some examples, after releasing the requested personal information to the assistant service, the assistant service is configured to use the personal information to fulfill a query submitted by the user. Additionally or alternatively, the assistant service may execute locally on the user device or on a server remote from the user device.

(6) In some implementations, the operations also include, in response to receiving the request for the personal information associated with the user, determining whether the requested personal information is stored in the centralized data store and determining, whether a set of user permissions associated with the user restrict the assistant service from accessing the requested personal information. In these implementations, when at least one of the requested personal information is not stored in the centralized data store or the set of user permissions associated with the user restrict the assistant service from accessing the requested personal information, the operations also include bypassing performance of the verification process and informing the assistant service that the user is required to provide the requested personal information in full to the assistant service. In these implementations, the operations may also include, after informing the assistant service that the user is required to provide the requested personal information in full: receiving, from the assistant service, a profile storage request requesting the profile service to store the requested personal information provided by the user to the assistant service; and storing the requested personal information provided by the user in the centralized data store. Here, the profile storage request includes the requested personal information provided in full by the user.

(7) Additionally or alternatively, in these implementations, the operations may also include, when the requested personal information is stored in the centralized data store and the set of user permissions associated with the user do not restrict the assistant service from accessing the requested personal information, determining whether the user device is suitable for receiving out-of-band notifications. Here, when the user device is suitable for receiving out-of-band notifications the operations also include: transmitting an out-of-band notification from the profile service to the user device, the out-of-band notification when received by the user device causing the user device to prompt the user to provide consent to release the requested personal information to the assistant service; receiving, from the user device, a consent response indicating that the user consents to releasing the requested personal information to the assistant service; and responsive to receiving the consent response, releasing, to the assistant service, the requested personal information stored on the centralized data store. The user device may be suitable for receiving out-of-band notifications when the user device includes a display screen and the user device may prompt the user to provide consent to release the requested personal information by displaying a selectable graphical element on the display screen, that when selected by the user causes the user device to transmit the consent response to the profile service. Moreover, determining whether the user device is suitable for receiving out-of-band notifications may be based on application settings associated with the assistant service indicating whether the profile service is restricted from communicating out-of-band notifications to the user device for gaining user consent for releasing the personal information. Optionally, the profile service may perform the verification process through the assistant service when the user device is not suitable for receiving out-of-band notifications.

(8) In some examples performing the verification process through the assistant service further includes generating an arbitrary phrase of one or more terms, wherein the generated arbitrary phrase includes the unique token. In these examples, instructing the assistant service to prompt the user to recite the unique token causes the assistant service to output a verification message from the user device that prompts the user to speak each term of the one or more terms of the arbitrary phrase to provide consent for releasing the personal information. In these examples, the verification message output from the user device may further indicate a type of the personal information the

assistant service is requesting the profile service to release.

(9) In some implementations performing the verification process through the assistant service further includes, extracting, from the requested personal information stored in the centralized data store, one or more personal information fragments, wherein the unique token includes the one or more fragments; and generating a security statement answered by the one or more personal information fragments of the unique token. In these implementations, instructing the assistant service to prompt the user to recite the unique token causes the assistant service to output the security statement from the user device for the user to answer via spoken input to provide consent for releasing the personal information, and processing the audio data to determine whether the transcription of the spoken utterance recites the unique token includes determining whether the transcription of the utterance recites the one or more fragments extracted from the requested personal information.

(10) The personal information may include demographic information associated with the user that includes at least one of an address, a phone number, payment information, a date of birth, a social security number, gender, contact information, marital status, or occupation. Additionally or alternatively, the personal information may include one or more user preferences that includes at least one of scheduling preferences, communication preferences, shipping preferences, music/movie genre(s), dietary preferences/restrictions, purchase history, contacts, or password(s).

(11) Another aspect of the disclosure provides a system that includes data processing hardware and memory hardware in communication with the data processing hardware. The memory hardware stores instructions that when executed on the data processing hardware causes the data processing hardware to perform operations that include receiving, at a profile service executing on the data processing hardware, from an assistant service interacting with a user device of a user, a request requesting the profile service to release personal information associated with the user to the assistant service. The operations also include performing, through the assistant service, a verification process to verify that the user consents to releasing the requested personal information to the assistant service by instructing the assistant service to prompt the user to recite a unique token prescribed to the user; receiving audio data characterizing a spoken utterance captured by the user device of the user, processing the audio data to determine whether a transcription of the spoken utterance recites the unique token, and when the transcription of the spoken utterance recites the unique token, releasing, to the assistant service, the requested personal information stored on a centralized data store managed by the profile service.

(12) This aspect may include one or more of the following optional features. In some implementations, the operations also include, while performing the verification process through the assistant service, processing the audio data to determine whether the spoken utterance captured by the user device was spoken by the user associated with the requested personal information. Here, releasing the requested personal information to the assistant service further includes releasing the requested personal information to the assistant service when the transcription of the spoken utterance recites the unique token and the utterance was spoken by the user.

(13) The personal information requested by the assistant service may pertain to personal information required by the assistant service in order to fulfill a query submitted by the user to the assistant service via the user device. In some examples, after releasing the requested personal information to the assistant service, the assistant service is configured to use the personal information to fulfill a query submitted by the user. Additionally or alternatively, the assistant service may execute locally on the user device or on a server remote from the user device.

(14) In some implementations, the operations also include, in response to receiving the request for the personal information associated with the user, determining whether the requested personal information is stored in the centralized data store and determining whether a set of user permissions associated with the user restrict the assistant service from accessing the requested personal information. In these implementations, when at least one of the requested personal information is

not stored in the centralized data store or the set of user permissions associated with the user restrict the assistant service from accessing the requested personal information, the operations also include bypassing performance of the verification process and informing the assistant service that the user is required to provide the requested personal information in full to the assistant service. In these implementations, the operations may also include, after informing the assistant service that the user is required to provide the requested personal information in full; receiving, from the assistant service, a profile storage request requesting the profile service to store the requested personal information provided by the user to the assistant service, and storing the requested personal information provided by the user in the centralized data store. Here, the profile storage request includes the requested personal information provided in full by the user.

(15) Additionally or alternatively, in these implementations, the operations may also include, when the requested personal information is stored in the centralized data store and the set of user permissions associated with the user do not restrict the assistant service from accessing the requested personal information, determining whether the user device is suitable for receiving out-of-band notifications. Here, when the user device is suitable for receiving out-of-band notifications the operations also include transmitting an out-of-band notification from the profile service to the user device, the out-of-band notification when received by the user device causing the user device to prompt the user to provide consent to release the requested personal information to the assistant service; receiving, from the user device, a consent response indicating that the user consents to releasing the requested personal information to the assistant service, and responsive to receiving the consent response, releasing, to the assistant service, the requested personal information stored on the centralized data store. The user device may be suitable for receiving out-of-band notifications when the user device includes a display screen and the user device may prompt the user to provide consent to release the requested personal information by displaying a selectable graphical element on the display screen, that when selected by the user causes the user device to transmit the consent response to the profile service. Moreover, determining whether the user device is suitable for receiving out-of-band notifications may be based on application settings associated with the assistant service indicating whether the profile service is restricted from communicating out-of-band notifications to the user-device for gaining user consent for releasing the personal information. Optionally, the profile service may perform the verification process through the assistant service when the user device is not suitable for receiving out-of-band notifications.

(16) In some examples, performing the verification process through the assistant service further includes generating an arbitrary phrase of one or more terms, wherein the generated arbitrary phrase includes the unique token. In these examples, instructing the assistant service to prompt the user to recite the unique token causes the assistant service to output a verification message from the user device that prompts the user to speak each term of the one or more terms of the arbitrary phrase to provide consent for releasing the personal information. In these examples, the verification message output from the user device may further indicate a type of the personal information the assistant service is requesting the profile service to release.

(17) In some implementations, performing the verification process through the assistant, service further includes, extracting, from the requested personal information stored in the centralized data store, one or more personal information fragments, wherein the unique token includes the one or more fragments; and generating a security statement answered by the one or more personal information fragments of the unique token. In these implementations, instructing the assistant service to prompt the user to recite the unique token causes the assistant service to output the security statement from the user device for the user to answer via spoken input to provide consent for releasing the personal information, and processing the audio data to determine whether the transcription of the spoken utterance recites the unique token includes determining whether the transcription of the utterance recites the one or more fragments extracted from the requested personal information.

(18) The personal information may include demographic information associated with the user that includes at least one of an address, a phone number, payment information, a date of birth, a social security number, gender, contact information, marital status, or occupation. Additionally or alternatively, the personal information may include one or more user preferences that includes at least one of scheduling preferences, communication preferences, shipping preferences, music/movie genre(s), dietary preferences/restrictions, purchase history, contacts, or password(s).

(19) The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description and drawings, and from the claims

Description

DESCRIPTION OF DRAWINGS

- (1) FIGS. 1A-1C are schematic views of an example speech environment for sharing personal information across assistant services.
- (2) FIG. 2 is a schematic view of example personal information **200** for a user profile stored in a centralized data store.
- (3) FIG. 2 is a schematic view of example components of an assistant service.
- (4) FIG. 4 is a schematic view of an example speech environment where a user is required to input personal information in full.
- (5) FIG. 5 is a schematic view of an example computing device that may be used to implement the systems and methods described herein.
- (6) FIG. 6 is a flowchart of an example arrangement of operations for a method of sharing assistant profiles verified via speaker identification.
- (7) Like reference symbols in the various drawings indicate like elements.

DETAILED DESCRIPTION

- (8) Users frequently interact with voice-enabled assistant interfaces on smart devices such as, without limitation, phones, watches, and smart speakers/displays. These assistant interfaces enable users to get things done and find answers to questions they might have, all through natural, conversational interactions. Developers are increasingly developing voice-enabled assistant interfaces for applications and services to permit user interaction through natural conversation. For example, automatic speech recognition (ASR) and natural language understanding (NLU) models may recognize and interpret queries spoken by users and fetch responses to these spoken queries.
- (9) As users begin to interact with many different assistant services, it can become an annoyance to require users to input common details of personal information repeatedly across the different assistant services yet still keep the personal information private/secure. As used herein, personal information can include demographic information such as phone numbers), home/work address(es), date of birth, payment details, and the like. Personal information may also include user-specific preferences that pertain to the user and permit assistants to interact with a user in a more tailored and personalized manner. For instance, user-specific preferences may include attributes that convey details about the user such as the user does not like to schedule appointments on a particular day of the week (e.g., Monday), dietary restrictions/preferences (e.g., the user is vegetarian and allergic to dairy), recent purchase history (e.g., the user recently purchased running shoes). The user-specific preferences can be input explicitly by the user and/or learned over time based on past, interactions with one or more assistant services.
- (10) Personal information pertaining to a particular user may be stored in a central location such as a centralized data store hosted by a cloud service provider and managed by a profile service of the cloud service provider. In these instances, the cloud service provider may host many applications/services used by the user and the user may explicitly grant permission to store the

personal information at the centralized data store as part of a user profile pertaining to the user. The profile service may manage storage and access to user profiles for a population of users that correspond to customers of the cloud service provider. These users/clients may explicitly grant permission to the profile service for storing their personal information and profiles in the centralized data store. Additionally, these users/clients may also define application permissions indicating which services/applications are allowed to access the stored personal information and/or which services/applications are restricted from accessing the stored personal information. As such, the profile service may grant, with the consent of the user, these applications/services access to the personal information stored in the centralized data store when needed without having to burden the user by requiring these applications/services to request the user to input the same personal information repeatedly. The profile service always provides the user the option to delete any of the personal information at the centralized data store and allow the user to revoke permission previously granted by the user to store the personal information at the central data store at any time.

(11) Implementations herein are directed toward techniques for supporting the sharing of personal information pertaining to a particular user across multiple assistant services in a private and secure manner to thereby eliminate any requirement by the user to re-enter (e.g., via text or speech) common attributes from the personal information when requested by the different assistant services. Specifically, the user may store the personal information in a centralized data store and explicitly grant consent to a profile service (e.g., hosted by a cloud service provider) to permit the assistant services to access the personal information stored in the centralized data store. Notably, an assistant service requesting retrieval of personal information from the centralized data store for a particular user may trigger a verification step that the user must perform in order to grant consent to the assistant service for the stored personal information in the centralized data store.

(12) As will become apparent, this verification step may leverage speech recognition and speaker identification techniques to verify utterances spoken by the user of unique tokens and/or fragments from the personal information requested by the assistant service. For example, when the assistant service is requesting the profile service for access to a street address of a user's residence stored in the centralized data store, the profile service may extract the user's house number (e.g., 5-1-0-5) and city (e.g., Atlantic City) from the user's street address as fragments and generate a security statement (e.g., "What is your house number and what city do you live in") answered by the fragments. To permit the user to provide consent for releasing the requested personal information, the profile service may instruct the assistant service to output the security statement from the user device for the user to answer via a spoken utterance. The security statement may be in the form of a question or as a command (e.g., "Please provide your house number and the city you live in"). Thereafter, the profile service processes audio data of the spoken utterance captured by the user device to provide two-factor authentication where a transcription of the utterance must recite the correct house number (e.g., 5-1-0-5) and city name (e.g., Atlantic City) and a speaker-discriminative vector extracted from the audio data must match a reference speaker-discriminative vector associated with the user. As such, the profile service may perform speech recognition on the audio data to generate a transcription of the utterance to ascertain the house number and city name spoken by the user and determine whether it matches the corresponding house number and city name stored in the data store for the user. At the same time, the profile service may execute a speaker verification process by performing text-independent speaker identification (TI-SID) on the audio data to extract the speaker-discriminative vector and determine whether it matches the reference speaker-discriminative vector stored in the user profile for the user in the centralized data store. In addition to or in lieu of prompting the user to speak a fragment of the requested information, the profile service may generate a unique token (e.g., "Bumblebee") and prompt the user to speak the unique token. In this scenario, the profile service may execute the speaker verification process by performing TI-SID on audio data characterizing the user uttering "Bumblebee" to extract the speaker-discriminative vector and determine whether it matches the

reference speaker-discriminative vector for the user. Additionally or alternatively, the unique token may be a password known to the user and in which a text-dependent reference speaker-discriminative vector exists for the user where the user spoke the password during an enrollment process.

(13) FIGS. 1A-1C show a speech environment **100** including a user **10** interacting with an assistant service **300** running on an assistant-enabled device (also referred to as user device **110**, device **110**, or an AED **110**). In the example shown, the assistant service **300** corresponds to a custom assistant service created by a developer to allow users to discover and discuss books, while also supporting a flow for purchasing a book. Notably, the user **10** may interact with other custom assistant services via the user device **110** and/or other devices, where these other custom assistants are also developed by developers for performing tasks in other domains of interest to the user **10**.

(14) FIG. 1A shows the assistant service **300** outputting synthesized speech **18** from the user device **110** as part of a dialogue where the assistant service **300** states “*Fundamentals of Meteorology* explains weather patterns really well”. In response, the user **10** speaks a query **20** directed toward the assistant service **300** that is captured by the user device **110** in streaming audio indicating that the user **10** would like to purchase the book *Fundamentals of Meteorology*. Thus, the query **20** refers to a request to perform an action, operation, or task, and more specifically, a request for the assistant service **300** to perform an action, operation, or task related to purchasing the book for the user **10**.

(15) In order to fulfill the query **20** (purchase and ship the book) spoken by the user **10**, the assistant service **300** needs personal information **200** pertaining to the user's **10** payment details (e.g., credit card information) and home address. Notably, the user **10** may interact with a multitude of other custom assistant services through the user device **102** or other devices, and on one or more previous occasions, some of these other assistant services may have required these same attributes of personal information (e.g., payment details and home address). For instance, the user **10** may also use a custom assistant service for booking trips where the user **10** provided payment details for reserving a hotel and purchasing a plane ticket and also provided his/her home address as a pick-up point for a cab for taking the user **10** to the airport. In this scenario, the payment details and home address provided by the user **10** while interacting with the trip booking assistant service may be stored (with the user's **10** consent) in centralized data storage (also referred to as centralized data store) that a profile service **170** manages for a population of users. As will become apparent, the assistant service **300** for discovering, discussing, and purchasing books may provide a personal information request **325** to the profile service **170** requesting release of the payment details and home address stored as personal information **200** in the centralized data storage **168**. The profile service **170** may verify the identity of the user and verify that the user **10** consents to releasing the personal information **200** to the assistant service **300** and then release the requested personal information **200** to the assistant service **300** to fulfill the query **20** without requiring the user **10** to input the personal information in full.

(16) Some examples of user devices **110** include, but are not limited to, mobile devices (e.g., mobile phones, tablets, laptops, e-book readers, etc.), computers, wearable devices (e.g., smart watches), music players, casting devices, smart appliances (e.g., smart televisions), vehicle infotainment devices, internet of things (IoT) devices, remote controls, smart speakers, etc. The user device **110** includes data processing hardware **111** and memory hardware **112** storing instructions that when executed on the data processing hardware **111** cause the data processing hardware **111** to perform operations. The user device **110** includes an array of one or more microphones configured to capture acoustic sounds such as speech directed toward the user device **110**. The user device **110** may also include, or be in communication with, an audio output device (e.g., speaker) configured to output audio such as synthesized speech **18**, **22** from the assistant service **300** and other assistant services the user **10** interacts with from time to time. As used herein, each assistant service **300** may also be referred to as a digital assistant or digital assistant interface.

(17) In the example of FIG. 1A, the user device **102** does not include a display, however, in other examples, FIG. 1C shows the user device **102** having a display **116** configured to display graphical user interface (GUI) elements (e.g., windows, screens, icons, menus, etc.) anchor graphical content. For example, the device **110** may load or launch applications that generate GUI elements or other graphical content for the display **116**. Moreover, the elements generated in the display **116** may be selectable by the user **10** and also serve to provide some form of visual feedback to processing activities and/or operations occurring on the device **110**. Furthermore, since the device **110** is a voice-enabled device **110**, the user **10** may interact with elements generated on the display **116** using various voice commands.

(18) In some implementations, the device **110** communicates via a network **130** with a remote system **160** (also referred to as a remote server **160**). The remote system may include remote resources **162**, such as remote data processing hardware **164** (e.g., remote servers or CPUs) and/or remote memory hardware **166** (e.g., remote databases or other storage hardware). The device **110** may utilize the remote resources **162** to perform various functionality related to conversational assistant services **300** the user **10** interacts with. For instance, some portion of the assistant service **300** may reside on the remote system **160** where a developer built the assistant service **300** as a custom, voice-based digital assistant using a set of cloud application programming interfaces (APIs) hosted by the remote system **160**. As such, the remote system **160** may correspond to a cloud service provider hosting a multitude of custom assistant services **300**. In one example, assistant service **300** executes locally on the device **110**. In another example, the assistant service **300** resides on the remote system **160** and communicates with the user device **110** over the network **130**. In yet another example, functionality of the assistant service **300** is split across the device **110** and the remote system **160** (e.g., the device **110** and the remote system **160** processes aspects of the assistant service **300** in parallel).

(19) The profile service **170** executes on the data processing hardware **164** of the remote system **160** and is configured to manage access to user profiles **12**, **12a-n** stored in the centralized data storage **168** overlain on the memory hardware **166**. Each user profile **12** is associated with a respective user who may interact with one or more assistant services hosted by the remote system **160**, other remote systems, locally on one or more user devices, or some combination thereof. Each user profile **12** includes, without limitation, personal information **200** associated with the respective user and may also include a set of user permissions **14**. FIG. 2 shows an example of personal information **200** in a respective user profile. For instance, the personal information **200** may include demographic information such as, without limitation, one or more addresses (e.g., home and/or work addresses), one or more phone numbers, payment information, date of birth, social security number, gender, contact information, marital status, and an occupation of the respective user. The personal information **200** may also include user preferences such as, without limitation, scheduling preferences (e.g., the respective user does not like to schedule appointments in the mornings or on Mondays), communication preferences (prefers to receive text messages), shipping preferences, music/movie genre(s) the respective user likes, dietary preferences/restrictions, purchase history, or passwords. Some of the user preferences may be explicitly input by the user, while other user preferences may be learned and updated based on interactions with assistant services and/or other services hosted by the remote system **160**.

(20) In response to receiving the request **325** from the assistant service **300** for the personal information **200**, FIG. 1A shows the profile service **170** determining whether the requested personal information **200** is stored in the centralized data store **168** and whether the set of user permissions **14** associated with the user **10** restrict the assistant service **300** from accessing the requested personal information **200**. Here, the personal information request **325** may identify the type of personal information the assistant service **300** needs as well as a user identifier identifying the user **10** so that the profile service **170** pulls the respective user profile **12** associated with the user **10** from the data storage **168** to inspect the personal information **200** and the user permissions

14. The personal information request **325** may also include a device identifier that uniquely identifies the user device **102** the user **10** is using to interact with the assistant service **300**. The device identifier may indicate, or be used to obtain, a type of the user device (e.g., phone, speaker, make/model, etc.), peripherals of the user device (e.g., display or no display?), an operating system running on the user device **110**, or other characteristics associated with the user device **110**.

(21) When at least one of the requested personal information **200** is not stored in the centralized data store **168** or the set of user permissions **14** associated with the user **10** restrict the assistant service **300** from accessing the requested personal information, the profile service **170** bypasses performance of a verification process depicted in FIGS. **1A-1C** for verifying that the user **10** consents to releasing the personal information **200** to the assistant service **300** and instead informs the assistant service **300** that the user **10** is required to provide the requested personal information **200** in full to the assistant service. For instance, FIG. **4** shows the profile service **170** responding to the assistant service **300** with denial instructions **402** that cause the assistant service **300** to prompt the user **10** to input the personal information **200** (e.g., payment information and address) needed by the assistant service **300** in order to fulfill the query **20** to purchase and ship the book to the user **10**. In some examples, the assistant service **300** prompts the user **10** by outputting synthesized speech **28** from the user device that requests the user to “Please input your shipping address and payment details in full so that I can ship you *Fundamentals of Meteorology*”. Subsequently, the user **10** inputs the personal information **200** (e.g., by spoken input or textual input) to the user device **110** for use by the assistant service **300** to fulfill the query **20**. In some examples, the assistant service **300** submits a storage request **410** to the profile service **170** that includes the personal information **200** input by the user **10** and requests the profile service **170** to store the personal information **200** (e.g., payment details and address) in a user profile **12** for the user **10** stored in the centralized data storage **168**. Notably, provided the user permissions **14** allow, these attributes of the personal information **200** stored in the centralized data storage **108** may be released to the assistant service **300** or other assistant services as needed so long as the user consents to the release of the personal information **200**.

(22) When the requested personal information **200** is stored in the centralized data store **168** and the set of user permissions **14** associated with the user **10** do not restrict the assistant service **300** from accessing the requested personal information, FIG. **1B** shows the profile service **170** performing the verification process through the assistant service **300**. Optionally, the profile service **170** may determine whether the user device **110** is suitable for receiving out-of-band (OOB) notifications directly from the profile service **170** and only perform the verification process through the assistant service **300** when the user device **110** is not suitable for receiving OOB notifications. In some examples, the user device is not suitable for receiving OOB notifications when the user device does not include a display screen and/or application settings associated with the assistant service **300** indicate that the assistant service **300** is restricted from communicating OOB notifications to the user device **110** for gaining user consent for releasing the personal information. An operating system the user device **110** runs may be ascertained from the device identifier contained in the personal information request **325** and be applied as an attribute for determining whether or not the user device **110** is suitable for receiving OOB notifications.

(23) When the profile service **170** determines the user device **110** is suitable for receiving OOB notifications (i.e., the user device **110** includes a display **116** and the application settings to not restrict OOB notifications), FIG. **1C** depicts the profile service **170** performing the verification process without the assistant service **300** by transmitting an OOB notification **250** to the user device **110** that causes the user device **110** to prompt the user **10** to provide consent to release the requested personal information to the assistant service **300**. Notably, the OOB notification **250** is communicated directly from the profile service **170** to the user device **110** via the network **130** without the need of the assistant service **300**. The user device **110** may display the prompt in a GUI presented on the display **116** and/or present the prompt via synthesized speech. In the example

shown, the prompt is displayed as a textual message on the display **116** and provides selectable graphical elements that permit the user **10** to provide a user input indication indicating selection of the “Yes” graphical element to provide consent for releasing the personal information **200**. Similarly, the user device **110** may receive a user input indication indicating selection of the “No” graphical element to deny consent for releasing the personal information. A timeout resulting in a denied consent may occur when the user **10** does not provide any selection within a predetermined period of time. When the user **10** provides the input indication indicating selection of the “Yes” graphical element displayed on the display **116**, the user device **110** transmits a consent response **252** to the profile service **170** indicating that the user **10** consents to releasing the requested personal information **200** to the assistant service **300**. Accordingly, the profile service **170** may release the requested personal information **200** stored on the centralized data store to the assistant service **300** in response to receiving the consent response **252**. Selection of the “No” graphical element or no selection before the timeout may result in the user device **110** transmitting a consent denial response indicating that the user does not consent to releasing the personal information **200** to the assistant service **300**.

(24) Referring back to FIG. **1B**, the profile service **170** performs the verification process through the assistant service **300** to verify that the user **10** consents to releasing the requested personal information **200** to the assistant service **300**. Here, the profile service **170** instructs the assistant service **300** to prompt the user to recite a unique token **174** prescribed to the user **10**. The profile service **170** includes a token generator **172** configured to generate the unique token **174** prescribed to the user **10**. In some examples, the token generator **172** generates an arbitrary phrase/string of one or more terms as the unique token **174** for the user **10** to recite. As the token **174** is uniquely generated for the user, it is robust for preventing replay attacks where someone may try to spoof the user by submitting offline copies of recordings of the user's voice. For instance, the token generator **172** may generate the arbitrary phrase “Bumblebee” as the unique token **174** prescribed to the user **10** to recite. As such, the instructions received by the assistant service **300** may cause the assistant service **300** to output a verification message as synthesized speech **22** from the user device **110** that prompts the user to speak each term of the one or more terms of the arbitrary phrase (e.g., Bumblebee) to provide consent for releasing the personal information **200**. Additionally, the verification message output from the user device **110** as synthesized speech **22** may further indicate the type of personal information the assistant service is requesting the profile service to release.

(25) In additional examples, the token generator **172** extracts, from the requested personal information **200** stored in the centralized data store **168**, one or more personal information fragments **202** for use as the unique token **174** for the user **10** to recite. In the example shown, the requested personal information pertains to payment details and an address so the token generator **172** may extract fragments **202** that include house number from the full address of the user **10**. For instance, the user's address may be 139 Stoddard Ave., Atlantic City, New Jersey and the fragments **202** extracted therefrom may include “**139**” that corresponds to the house number portion of the address. In these examples, the profile service **170** further generates a security statement **175** answered by the personal fragments **202** of the unique token **174**, whereby the instructions received by the assistant service **300** may cause the assistant service to output the synthesized speech **22** conveying the security statement **175** from the user device **110** for the user **10** to answer via spoken input to provide consent for releasing the personal information.

(26) In the example shown, the verification message output from the user device **110** as synthesized speech **22** states “So that I can access your address and payment details, please say ‘Bumblebee’ and speak your house number”. Notably, the unique token **174** corresponding to the arbitrary phrase is explicitly conveyed in the verification message, while the security statement **175** is conveyed in the verification message as a challenge for the user **10** to speak the correct house number as the unique token **174** including the personal information fragments **202**. To alleviate the burden from having to manually input the address and payment details in full, the user **10**

acknowledges the verification message and provides consent for release of the payment information by speaking the utterance **24** “Bumblebee 1-3-9”. The user device **110** captures the spoken utterance **24** and transmits audio data **121** characterizing the spoken utterance **24** to the profile service **170** for verification. Here, the profile service **170** executes an automatic speech recognition (ASR) system/model **176** for processing the received audio data **121** to generate a transcription **178** and executes a verifier **180** to determine whether the transcription **178** of the spoken utterance **24** recites each unique token **174** generated by the token generator **172**. Continuing with the example, the verifier **180** is configured to verify the user consent when the transcription **178** recites both the unique token **174** corresponding the arbitrary phrase “Bumblebee” and the unique token **174** corresponding to the fragments **202** associated with the house number “1-3-9” of the user's address. As such, the profile service **170** may release the requested personal information **200** (e.g., the user's address and payment details) when the verifier determines the transcription **178** recites the unique tokens **174**.

(27) In some implementations, the profile service **170** processes the received audio data **121** to determine whether the spoken utterance **24** captured by the user device **110** was spoken by the user **10** associated with the requested personal information and only releases the personal information **200** to the assistant service **300** when both the verifier **180** determines the transcription **178** recites the unique tokens **174** and the utterance **24** was spoken by the user **10** and not some other user. In these implementations, the profile service **170** may execute a speaker verification (SV) model **184** configured to receive the audio data **121** as input and generate, as output, an evaluation vector **186** representing voice characteristics of the spoken utterance **24**. A scorer **188** may determine a SV confidence score indicating a probability/likelihood of the evaluation vector **186** matching a reference vector **205** for the user **10**. The reference vector **205** may be obtained from the user profile **12** associated with the user and stored in the centralized data store **168**. The SV model **184** may generate the reference vector **205** for the user **10** during a voice enrollment process where the reference vector **205** represents characteristics of the voice of the user. As such, the profile service **170** may verify the identity of the user **10** as the speaker of the spoken utterance **24** when the SV confidence score satisfies a confidence threshold. Notably, the user device **110** may support multiple enrolled users **10** each having a respective reference vector **205** generated by the SV model **184** during a voice enrollment process for the respective user **10**. For instance, multiple family members may use a smart speaker residing in the family's home where each family member can undergo the voice enrollment process with the SV model **184** to generate a respective reference vector **205** for that family member that may be stored in the central data store **168**.

(28) The speaker verification model **134** may include a text-independent speaker verification (TI-SV) model **184** for generating text-independent (TI) evaluation vectors **186**. Notably, the TI-SV model **184** is beneficial since the unique token(s) **174** the user is reciting in the spoken utterance are not predetermined or known a priori. However, the speaker verification model **184** may include a text-dependent speaker verification (TD-SV) model **184** for generating text-dependent (TD) evaluation vectors **186** representing voice characteristics of a user speaking a unique token **174**. In this scenario, the unique token **174** may include a password/passphrase only known to the user and for which a TD reference vector **205** is obtained during a voice enrollment process of the user speaking the predetermined password/passphrase. In some scenarios, the TD-SV model **184** is generated dynamically using past instances of the user speaking a unique token as training data. These past instances could pertain to previous interactions with the given assistant service **300** or other assistant services.

(29) In the example shown, the token generator **172** effectively generates two unique tokens **174** for the user **10** to recite that includes the arbitrary phrase “Bumblebee” and the personal information fragments “**139**” corresponding to the house number extracted from the user's address. The profile service **70** may only generate one of these unique tokens **174** for the user **10** to recite in order to validate the user's consent or the profile service **170** may generate more than two unique

tokens **174** for validation. A level of sensitivity of the personal information **200** requested may impact the type, complexity, and number of unique tokens **174** the user needs to recite in order to verify the user consent. Depending on the level of sensitivity of the personal information requested (e.g., phone number versus credit card details), the profile service **170** may prompt the assistant service **300** to obtain additional verification signals, such as biometric markers including finger/palm print, face/retina ID, etc.

(30) With continued reference to FIG. **1B**, when the transcription **178** of the spoken utterance **24** recites (permitting some leeway) the unique tokens and the scorer **188** determines the utterance **24** was spoken by the user (e.g., the identity of the user is verified as the speaker of the utterance), the profile service verifies that the user **10** consents to the release of the requested personal information and thereby releases the requested personal information **200** stored on the centralized data store **168** to the assistants service **300**. The profile service **170** may remember that the user **10** consented to the release of the personal information **200** to the assistant service **300** and permit the assistant service to re-access the same personal information when needed in the future without performing the verification process or requiring a weaker challenge to the user for re-access. If the personal information changes (e.g., the user moves to a new address), the user profile **12** stored in the centralized data store **168** may be updated to include the new address.

(31) The ASR system/model **176** and/or SV model **184** may execute on the remote system or on the user device **110**. Optionally, the profile service **170** may leverage a speech recognizer **310** (FIG. **3**) of the assistant service **300** to generate the transcription **178** of the spoken utterance **24**.

(32) When the verification process determines that the transcription **178** does not recite the unique tokens and/or the scorer **188** determines the utterance **24** was not spoken by the user (e.g. the identity of the user cannot be verified as the speaker of the utterance), the profile service **170** may instruct the assistant service **300** to prompt the user **10** to speak the unique token **174** a second time or simply inform the assistant service **300** to obtain the requested personal information in full via manual user input.

(33) FIG. **3** shows a schematic view of an example assistant service **300** which generally includes a speech recognizer **310**, an interpreter **320**, and an executor **330**. The speech recognizer **310** receives, as input, the audio data **120** characterizing the query **20** spoken by the user **10** that includes “That sounds perfect, I'd like to purchase this book” shown in FIG. **1A** and generates, as output, a speech recognition result **312** (e.g., transcription) of the query **20**. The speech recognizer **310** may include an end-to-end speech recognition model including a plurality of neural network layers. The speech recognizer **310** may include a conventional speech recognition system having acoustic, pronunciation, and language models.

(34) The interpreter **220** receives the speech recognition result **312** and may perform semantic interpretation (e.g., grammar interpretation) on the speech recognition result **312** to understand a context of the query **20** in order to identify an action **322** to perform in order to fulfil the query. Continuing with the example, the action **322** includes completing a transaction for the purchase of the book *Fundamentals of Meteorology* and shipment of the book to the user's address. Notably, the interpreter **320** learns that the personal information **200** pertaining to the user's payment details and address are needed for performing the action **322**. As such, the interpreter **320** may generate the personal information request **325** that requests the profile service **170** to release the personal information **200**. Using the techniques described above, the profile service **174** may release the requested personal information **200** pertaining to the payment details and address of the user to the executor **330**. The executor **330** may be configured to perform the action **322** by purchasing the book from a merchant using the payment details of the user and brokering the shipment of the book to the user's address.

(35) A software application (i.e., a software resource) may refer to computer software that causes a computing device to perform a task. In some examples, a software application may be referred to as an “application,” an “app,” or a “program.” Example applications include, but are not limited to,

system diagnostic applications, system management applications, system maintenance applications, word processing applications, spreadsheet applications, messaging applications, media streaming applications, social networking applications, and gaming applications.

(36) The non-transitory memory may be physical devices used to store programs (e.g., sequences of instructions) or data (e.g., program state information) on a temporary or permanent basis for use by a computing device. The non-transitory memory may be volatile and/or non-volatile addressable semiconductor memory. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(37) FIG. 5 is schematic view of an example computing device **500** that may be used to implement the systems and methods described in this document. The computing device **500** is intended to represent various forms of digital computers, such as laptops, desktops, workstations, personal digital assistants, servers, blade servers, mainframes, and other appropriate computers. The components shown here, their connections and relationships, and their functions, are meant to be exemplary only, and are not meant to limit implementations of the inventions described and/or claimed in this document.

(38) The computing device **500** includes a processor **510**, memory **520**, a storage device **530**, a high-speed interface/controller **540** connecting to the memory **520** and high-speed expansion ports **550**, and a low speed interface/controller **560** connecting to a low speed bus **570** and a storage device **530**. Each of the components **510**, **520**, **530**, **540**, **550**, and **560**, are interconnected using various busses, and may be mounted on a common motherboard or in other manners as appropriate. The processor **510** can process instructions for execution within the computing device **500**, including instructions stored in the memory **520** or on the storage device **530** to display graphical information for a graphical user interface (GUI) on an external input/output device, such as display **580** coupled to high speed interface **540**. In other implementations, multiple processors and/or multiple buses may be used, as appropriate, along with multiple memories and types of memory. Also, multiple computing devices **500** may be connected, with each device providing portions of the necessary operations (e.g., as a server bank, a group of blade servers, or a multi-processor system).

(39) The memory **520** stores information non-transitorily within the computing device **500**. The memory **520** may be a computer-readable medium, a volatile memory unit(s), or non-volatile memory unit(s). The non-transitory memory **520** may be physical devices used to store programs (e.g., sequences of instructions; or data (e.g., program state information; on a temporary or permanent basis for use by the computing device **500**. Examples of non-volatile memory include, but are not limited to, flash memory and read-only memory (ROM)/programmable read-only memory (PROM)/erasable programmable read-only memory (EPROM)/electronically erasable programmable read-only memory (EEPROM) (e.g., typically used for firmware, such as boot programs). Examples of volatile memory include, but are not limited to, random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), phase change memory (PCM) as well as disks or tapes.

(40) The storage device **530** is capable of providing mass storage for the computing device **500**. In some implementations, the storage device **530** is a computer-readable medium. In various different implementations, the storage device **530** may be a floppy disk device, a hard disk device, an optical disk device, or a tape device, a flash memory or other similar solid state memory device, or an array of devices, including devices in a storage area network or other configurations. In additional implementations, a computer program product is tangibly embodied in an information carrier. The

computer program product contains instructions that, when executed, perform one or more methods, such as those described above. The information carrier is a computer- or machine-readable medium, such as the memory **520**, the storage device **530**, or memory on processor **510**.

(41) The high speed controller **540** manages bandwidth-intensive operations for the computing device **500**, while the low speed controller **560** manages lower bandwidth-intensive operations. Such allocation of duties is exemplary only. In some implementations, the high-speed controller **540** is coupled to the memory **520**, the display **580** (e.g., through a graphics processor or accelerator), and to the high-speed expansion ports **550**, which may accept various expansion cards (not shown). In some implementations, the low-speed controller **560** is coupled to the storage device **530** and a low-speed expansion port **590**. The low-speed expansion port **590**, which may include various communication ports (e.g., USB, Bluetooth, Ethernet, wireless Ethernet), may be coupled to one or more input/output devices, such as a keyboard, a pointing device, a scanner, or a networking device such as a switch or router, e.g. through a network adapter.

(42) The computing device **500** may be implemented in a number of different forms, as shown in the figure. For example, it may be implemented as a standard server **500a** or multiple times in a group of such servers **500a**, as a laptop computer **500b**, or as part of a rack server system **500c**.

(43) FIG. **6** is a flowchart of an example arrangement of operations for a method **600** of sharing assistant profiles verified via speaker identification. The method **600** may execute on the data processing hardware **164** of the remote system **160** using instructions stored on the memory hardware **166** of the remote system **160**. At operation **602**, the method **600** includes receiving, at a profile service **170** executing on the data processing hardware **164**, from an assistant service **300** interacting with a user device **110** of a user **10**, a request requesting the profile service **170** to release personal information **200** associated with the user **10** to the assistant service **300**.

(44) At operation **604**, the method **600** includes performing, through the assistant service **300**, a verification process to verify that the user **10** consents to releasing the requested personal information **200** to the assistant service **300** by instructing the assistant service **300** to prompt the user **10** **174** to recite a unique token **174** prescribed to the user. At operation **606**, the method **600** further performs the verification process by receiving audio data **121** characterizing a spoken utterance **24** captured by the user device **110** of the user **10**.

(45) At operation **608**, the method **600** further performs the verification process by processing the audio data **121** to determine whether a transcription **176** of the spoken utterance **24** recites the unique token **174**. At operation **610**, the method **600** further performs the verification process by releasing, to the assistant service **300**, the requested personal information **200** stored on a centralized data store **168** managed by the profile service **170** when the transcription **176** of the spoken utterance **24** recites the unique token **174**.

(46) Various implementations of the systems and techniques described herein can be realized in digital electronic and/or optical circuitry, integrated circuitry, specially designed ASICs (application specific integrated circuits), computer hardware, firmware, software, and/or combinations thereof. These various implementations can include implementation in one or more computer programs that are executable and/or interpretable on a programmable system including at least one programmable processor, which may be special or general purpose, coupled to receive data and instructions from, and to transmit data and instructions to, a storage system, at least one input device, and at least one output device.

(47) These computer programs (also known as programs, software, software applications or code) include machine instructions for a programmable processor, and can be implemented in a high-level procedural and/or object-oriented programming language, and/or in assembly/machine language. As used herein, the terms “machine-readable medium” and “computer-readable medium” refer to any computer program product, non-transitory computer readable medium, apparatus and/or device (e.g., magnetic discs, optical disks, memory. Programmable Logic Devices (PLDs)) used to provide machine instructions and/or data to a programmable processor, including a

machine-readable medium that receives machine instructions as a machine-readable signal. The term “machine-readable signal” refers to any signal used to provide machine instructions and/or data to a programmable processor.

(48) The processes and logic flows described in this specification can be performed by one or more programmable processors, also referred to as data processing hardware, executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by special purpose logic circuitry, e.g., an FPGA (field programmable gate array) or an ASIC (application specific integrated circuit). Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., EPROM, EEPROM, and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks, magneto optical disks, and CD ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(49) To provide for interaction with a user, one or more aspects of the disclosure can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display) monitor, or touch screen for displaying information to the user and optionally a keyboard and a pointing device, e.g., a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input, in addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a users client device in response to requests received from the web browser.

(50) A number of implementations have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the disclosure. Accordingly, other implementations are within the scope of the following claims.

Claims

1. A computer-implemented method when executed on data processing hardware causes the data processing hardware to perform operations comprising: receiving, at a profile service executing on the data processing hardware, from an assistant service executing on a user device of a user, a request requesting the profile service to release personal information associated with the user to the assistant service, wherein the assistant service sends the request to the profile service in response to the assistant service: receiving initial audio data characterizing a query spoken by the user that requests the assistant service to perform an action on behalf of the user; and processing the initial audio data to identify the personal information associated with the user as personal information explicitly stored by the user on a centralized data store managed by the profile service that the assistant service requires in order to perform the action on the behalf of the user; determining that the user device is not suitable for receiving out-of-band notifications directly from the profile service without the assistant service for gaining user consent for releasing the requested personal

information to the assistant service; and based on determining that the user device is not suitable for receiving the out-of-band notifications directly from the profile service, performing, through the assistant service, a verification process to verify that the user consents to releasing the requested personal information to the assistant service by: extracting, from the requested personal information explicitly stored by the user on the centralized data store that the assistant service requires in order to perform the action on the behalf of the user, one or more personal information fragments, the extracted one or more personal information fragments comprising a unique token prescribed to the user by the profile service; generating a security statement answered by the extracted one or more personal information fragments that comprise the unique token; generating a verification message that prompts the user to provide a spoken input that answers the security statement; instructing the assistant service to output the verification message from the user device; receiving audio data characterizing a spoken utterance captured by the user device of the user; processing the audio data to determine whether a transcription of the spoken utterance recites each of the one or more extracted personal information fragments; and when the transcription of the spoken utterance recites each of the one or more extracted personal information fragments, releasing, to the assistant service, the requested personal information stored on the centralized data store managed by the profile service, the requested personal information enabling the assistant service to perform the action on behalf of the user.

2. The computer-implemented method of claim 1, wherein the operations further comprise, while performing the verification process through the assistant service: processing the audio data to determine whether the spoken utterance captured by the user device was spoken by the user associated with the requested personal information, wherein releasing the requested personal information to the assistant service further comprises releasing the requested personal information to the assistant service when the transcription of the spoken utterance recites each of the one or more extracted personal information fragments and the utterance was spoken by the user.

3. The computer-implemented method of claim 1, wherein, after releasing the requested personal information to the assistant service, the assistant service is configured to use the personal information to perform the action on the behalf of the user.

4. The computer-implemented method of claim 1, wherein the assistant service executes on the user device and a server remote from the user device in parallel.

5. The computer-implemented method of claim 1, wherein the operations further comprise, in response to receiving the request for the personal information associated with the user: determining whether the requested personal information is stored in the centralized data store; determining whether a set of user permissions associated with the user restrict the assistant service from accessing the requested personal information; and when at least one of the requested personal information is not stored in the centralized data store or the set of user permissions associated with the user restrict the assistant service from accessing the requested personal information: bypassing performance of the verification process; and informing the assistant service that the user is required to provide the requested personal information in full to the assistant service.

6. The computer-implemented method of claim 5, wherein the operations further comprise, after informing the assistant service that the user is required to provide the requested personal information in full: receiving, from the assistant service, a profile storage request requesting the profile service to store the requested personal information provided by the user to the assistant service, the profile storage request comprising the requested personal information provided in full by the user; and storing the requested personal information provided by the user in the centralized data store.

7. The computer-implemented method of claim 1, wherein the user device is suitable for receiving out-of-band notifications when the user device comprises a display screen.

8. The computer-implemented method of claim 1, wherein determining the user device is not suitable for receiving the out-of-band notifications is based on application settings associated with

the assistant service indicating whether the profile service is restricted from communicating the out-of-band notifications directly to the user device for gaining user consent for releasing the personal information.

9. The computer-implemented method of claim 1, wherein performing the verification process through the assistant service further comprises: determining a level of sensitivity of the requested personal information; and based on the level of sensitivity of the requested personal information, generating an arbitrary phrase of one or more terms, the generated arbitrary phrase comprising the unique token, wherein: generating the verification message that prompts the user to provide the spoken input that answers the security statement comprises generating the verification message that prompts the user to provide the spoken input that recites the unique token and that answers the security statement; processing the audio data to determine whether the transcription of the spoken utterance recites each of the one or more extracted personal information fragments further comprises processing the audio data to determine whether the transcription of the spoken utterance recites each term of the one or more terms of the arbitrary phrase; and releasing the personal information stored on the centralized data store managed by the profile service when the transcription of the utterance recites each term of the one or more terms of the arbitrary phrase and each of the one or more extracted personal information fragments.

10. The computer-implemented method of claim 9, wherein the verification message output from the user device further indicates a type of the personal information the assistant service is requesting the profile service to release.

11. The computer-implemented method of claim 1, wherein the personal information comprises demographic information associated with the user that includes at least one of: an address; a phone number; payment information; a date of birth; social security number; gender; contact information; marital status; or occupation.

12. The computer-implemented method of claim 1, wherein the personal information comprises one or more user preferences comprising at least one of: scheduling preferences; communication preferences; shipping preferences; music/movie genre(s); dietary preferences/restrictions; purchase history; contacts; or password(s).

13. A system comprising: data processing hardware; and memory hardware in communication with the data processing hardware, the memory hardware storing instructions that when executed on the data processing hardware cause the data processing hardware to perform operations comprising, receiving, at a profile service executing on the data processing hardware, from an assistant service executing on a user device of a user, a request requesting the profile service to release personal information associated with the user to the assistant service, wherein the assistant service sends the request to the profile service in response to the assistant service: receiving initial audio data characterizing a query spoken by the user that requests the assistant service to perform an action on behalf of the user; and processing the initial audio data to identify the personal information associated with the user as personal information explicitly stored by the user on a centralized data store managed by the profile service that the assistant service requires in order to perform the action on the behalf of the user; determining that the user device is not suitable for receiving out-of-band notifications directly from the profile service without the assistant service for gaining user consent for releasing the requested personal information to the assistant service; and based on determining that the user device is not suitable for receiving the out-of-band notifications directly from the profile service, performing, through the assistant service, a verification process to verify that the user consents to releasing the requested personal information to the assistant service by: extracting, from the requested personal information explicitly stored by the user on the centralized data store that the assistant service requires in order to perform the action on the behalf of the user, one or more personal information fragments, the extracted one or more personal information fragments comprising a unique token prescribed to the user by the profile service; generating a security statement answered by the extracted one or more personal information fragments that

comprise the unique token; generating a verification message that prompts the user to provide a spoken input that answers the security statement; instructing the assistant service to output the verification message from the user device; receiving audio data characterizing a spoken utterance captured by the user device of the user; processing the audio data to determine whether a transcription of the spoken utterance recites each of the one or more extracted personal information fragments; and when the transcription of the spoken utterance recites each of the one or more extracted personal information fragments, releasing, to the assistant service, the requested personal information stored on the centralized data store managed by the profile service, the requested personal information enabling the assistant service to perform the action on behalf of the user.

14. The system of claim 13, wherein the operations further comprise, while performing the verification process through the assistant service: processing the audio data to determine whether the spoken utterance captured by the user device was spoken by the user associated with the requested personal information, wherein releasing the requested personal information to the assistant service further comprises releasing the requested personal information to the assistant service when the transcription of the spoken utterance recites each of the one or more extracted personal information fragments and the utterance was spoken by the user.

15. The system of claim 13, wherein, after releasing the requested personal information to the assistant service, the assistant service is configured to use the personal information to perform the action on the behalf of the user.

16. The system of claim 13, wherein the assistant service executes on the user device and a server remote from the user device in parallel.

17. The system of claim 13, wherein the operations further comprise, in response to receiving the request for the personal information associated with the user: determining whether the requested personal information is stored in the centralized data store; determining whether a set of user permissions associated with the user restrict the assistant service from accessing the requested personal information; and when at least one of the requested personal information is not stored in the centralized data store or the set of user permissions associated with the user restrict the assistant service from accessing the requested personal information: bypassing performance of the verification process; and informing the assistant service that the user is required to provide the requested personal information in full to the assistant service.

18. The system of claim 17, wherein the operations further comprise, after informing the assistant service that the user is required to provide the requested personal information in full: receiving, from the assistant service, a profile storage request requesting the profile service to store the requested personal information provided by the user to the assistant service, the profile storage request comprising the requested personal information provided in full by the user; and storing the requested personal information provided by the user in the centralized data store.

19. The system of claim 13, wherein the user device is suitable for receiving out-of-band notifications when the user device comprises a display screen.

20. The system of claim 13, wherein determining the user device is not suitable for receiving the out-of-band notifications is based on application settings associated with the assistant service indicating the profile service is restricted from communicating the out-of-band notifications directly to the user device for gaining user consent for releasing the personal information.

21. The system of claim 13, wherein performing the verification process through the assistant service further comprises: determining a level of sensitivity of the requested personal information; and based on the level of sensitivity of the requested personal information, generating an arbitrary phrase of one or more terms, the generated arbitrary phrase comprising the unique token, wherein: generating the verification message that prompts the user to provide the spoken input that answers the security statement comprises generating the verification message that prompts the user to provide the spoken input that recites the unique token and that answers the security statement; processing the audio data to determine whether the transcription of the spoken utterance recites

each of the one or more extracted personal information fragments further comprises processing the audio data to determine whether the transcription of the spoken utterance recites each term of the one or more terms of the arbitrary phrase; and releasing the personal information stored on the centralized data store managed by the profile service when the transcription of the utterance recites each term of the one or more terms of the arbitrary phrase and each of the one or more extracted personal information fragments.

22. The system of claim 21, wherein the verification message output from the user device further indicates a type of the personal information the assistant service is requesting the profile service to release.

23. The system of claim 13, wherein the personal information comprises demographic information associated with the user that includes at least one of: an address; a phone number; payment information; a date of birth; social security number; gender; contact information; marital status; or occupation.

24. The system of claim 13, wherein the personal information comprises one or more user preferences comprising at least one of: scheduling preferences; communication preferences; shipping preferences; music/movie genre(s); dietary preferences/restrictions; purchase history; contacts; or password(s).
